

Principal Component Regression (PCR)

Collinearity, Regularization View, Model Selection, and Diagnostics
Question Set

TTK4260: Multivariate Data Analysis and ML

A. Motivation and Core Idea

Question 1 (*Why do we need PCR? Collinearity*)

[4 points] Consider a multiple linear regression (MLR) model $y = \mathbf{X}\boldsymbol{\theta} + \varepsilon$ with mean-centered columns in $\mathbf{X}$. Explain (i) what it means that the predictors are *collinear*, and (ii) why collinearity can make the ordinary least squares (OLS) estimate $\hat{\boldsymbol{\theta}}_{\text{OLS}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$ numerically unstable and have high variance.

Question 2 (*PCR pipeline*)

[4 points] Write the three main steps of Principal Component Regression (PCR) for a single response variable y . Assume $\mathbf{X}$ is mean-centered (and possibly scaled). Use PCA notation $\mathbf{X} = \mathbf{T}\mathbf{P}^\top$ and let $\mathbf{T}_r$ denote the first r score columns.

Question 3 (*PCR vs OLS in matrix form*)

[6 points] Show that the PCR fitted values can be written as

$$\hat{\mathbf{y}}_{\text{PCR}} = \mathbf{X}\mathbf{P}_r(\mathbf{T}_r^\top \mathbf{T}_r)^{-1} \mathbf{T}_r^\top \mathbf{y}.$$

Compare this with the OLS projection matrix $\mathbf{H}_{\text{OLS}} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$ and explain the *key difference*.

Question 4 (*Why each β_j is a simple scalar*)

[5 points] Let $\mathbf{t}_j$ be the j -th score vector (column of $\mathbf{T}$) and suppose the scores are orthogonal. Show that the OLS coefficient for regressing y on the first r scores satisfies

$$\hat{\beta}_j = \frac{\mathbf{t}_j^\top \mathbf{y}}{\mathbf{t}_j^\top \mathbf{t}_j}, \quad j = 1, \dots, r.$$

Interpret this geometrically.

Question 5 (*Back-projection to original variables*)

[5 points] Suppose $\mathbf{X}$ is mean-centered and $\mathbf{T}_r = \mathbf{X}\mathbf{P}_r$. Show that the regression coefficients in the original variable space are

$$\hat{\boldsymbol{\theta}}_{\text{PCR}} = \mathbf{P}_r \hat{\boldsymbol{\beta}}.$$

If y was also mean-centered, how do you write the final prediction model for an uncentered new sample $\mathbf{x}_{\text{new}}$?

B. Geometry and Regularization View

Question 6 (*Project, then regress*)

[4 points] Describe the geometric interpretation of PCR in predictor space. Why can PCR reduce variance compared to OLS?

Question 7 (*PCR as shrinkage / hard thresholding*)

[6 points] Assume $\mathbf{X}$ has SVD $\mathbf{X} = \mathbf{U}\mathbf{S}\mathbf{V}^\top$ with singular values $s_1 \geq \dots \geq s_p$ and right singular vectors $\mathbf{v}_j$. Write the OLS solution in this basis and explain how PCR modifies it when keeping r components.

Question 8 (*Bias-variance tradeoff*)

[5 points] Explain qualitatively how the number of retained PCs r affects bias and variance in PCR. What happens at the extremes $r = 0$ and $r = p$ (full rank)?

Question 9 (*Head-to-head: PCR vs ridge*)

[6 points] PCR is often compared to ridge regression. State one similarity and one key difference between PCR and ridge regression in terms of shrinkage in the principal-component basis.

C. Model Selection and Cross-Validation

Question 10 (*Selecting r by cross-validation*)

[6 points] Describe a practical procedure to choose the number of components r using K -fold cross-validation (CV). What performance metric would you use for a regression problem?

Question 11 (*Avoiding data leakage*)

[5 points] In PCR with CV, why is it incorrect to compute PCA loadings $\mathbf{P}$ once on the full dataset and reuse them inside each fold? What is the correct approach?

Question 12 (*Explained variance vs predictive power*)

[6 points] You find that the first two PCs explain 90% of the variance in $\mathbf{X}$, but using only $r = 2$ components gives poor prediction of y . Give two reasons why high explained variance in $\mathbf{X}$ does not guarantee good predictive performance for y .

D. Diagnostics and Interpretation

Question 13 (*Predicted vs observed*)

[4 points] You have a fitted PCR model and compute predictions on a test set. What patterns in a predicted-vs-observed plot indicate (i) a good model, (ii) systematic bias, and (iii) heteroscedasticity or model mismatch?

Question 14 (*Residuals vs fitted*)

[5 points] What do you look for in a residuals-vs-fitted plot for a PCR model? Name two warning signs and what they might imply.

Question 15 (*Coefficient paths*)

[6 points] Define what is meant by a *PCR coefficient path* as a function of the number of retained components r . How can coefficient paths help interpret collinearity and model stability?

Question 16 (*Interpreting PCR in an application*)

[6 points] In a process setting (e.g., many correlated sensors $\rightarrow$ product quality y), you fit PCR with a small r and achieve good prediction. Give two ways PCR can be used for *insight* beyond prediction.

E. Limitations and Pitfalls

Question 17 (*When PCR can fail*)

[6 points] Give an example scenario (conceptual, not necessarily numeric) where PCR can perform poorly even though it reduces collinearity problems. What alternative method is often introduced to address this limitation?

Question 18 (*Centering and scaling choices*)

[5 points] In PCR, why is mean-centering of $\mathbf{X}$ almost always required? When is scaling (standardization) important, and what can go wrong if you skip it?

Question 19 (*Reporting a “full PCR story”*)

[6 points] List four items you would include when reporting PCR results in a scientific/engineering setting to tell the “full story” (not just a single RMSE number).